Post Machina: Autonomous Machines and Their Organizational Implications

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Abstract

The division of labor is an important determinant of worker productivity, yet little is known about how firms reorganize jobs when automation disrupts the set of tasks performed by workers. I extend the traditional tradeoff between specialization and coordination costs by modeling automation as a change to the content of work. The model predicts that automation changes worker specialization according to whether adoption creates new coordination demands or substitutes for coordination activities. I test these predictions using data from more than 119 million job postings and worker employment histories from more than 190,000 U.S. firms. Using a shift-share instrument for automation adoption, I find substantial heterogeneity across technologies: robotics and software broaden jobs, whereas AI narrows them. These differences are mirrored by changes in coordination work: robotics and software increase both coordination among workers and interaction between workers and technology, while AI reduces both. Technologies also differ in how they change the tacit content of human work. Robotics and AI leave workers performing relatively more tacit tasks, whereas software shifts their responsibilities toward more codified tasks. Together, these results show that automation changes the design of jobs, the coordination required within firms, and the tasks that remain with workers.

1 Introduction

The division of labor describes how firms allocate production tasks across workers. It is a central dimension of organizational design and an important determinant of productivity. Smith (1776) first showed that specialization raises output, using the production of pins to illustrate how dividing a production process into distinct operations allows workers to perform their work more quickly. Becker and Murphy (1992) later formalized why the division of labor has limits: workers become more productive as they specialize, but firms incur higher coordination costs as tasks are divided across a larger workforce, so a firm's division of labor is chosen only up to the point where these two forces balance. This tradeoff between the productivity gains from specialization and the costs of coordinating a more divided workforce remains one of the leading explanations for why firms organize production the way they do.

While this literature explains how firms distribute work across workers, research on automation has primarily examined which parts of that work are displaced by machines. Autor et al. (2003) introduce the task-based framework, showing that automation substitutes for some tasks within an occupation while leaving others to workers. Acemoglu and Restrepo (2018, 2019) formalize this displacement channel and show that automation can also reinstate workers into the new tasks it creates. Autor (2015) documents which kinds of work, tacit judgment in particular, have so far resisted automation. This body of research is detailed about which tasks and jobs automation eliminates, but it says little about the fate of the jobs that survive. This paper studies how firms redivide the surviving work into roles once some tasks have been automated away.

I measure firms' division of labor using job breadth: the number of tasks assigned to a worker. Broader jobs contain more tasks per worker and therefore correspond to less specialization, while narrower jobs contain fewer tasks and correspond to greater specialization. The central question is whether changes in the human task set induced by automation lead firms to reorganize the boundaries of jobs, and why different automation technologies may generate different organizational responses.

I extend the specialization-coordination tradeoff from Becker and Murphy (1992) while allowing automation to change the codified content of work. Tasks differ in codifiability. Codified work relies on knowledge that can be readily articulated and transferred, whereas tacit work relies more heavily on knowledge that is difficult to communicate completely. Because tacit knowledge is lost when work must be coordinated across worker boundaries, tacit tasks are more costly to divide among highly specialized workers. At the same time, codified tasks receive greater productivity benefits from specialization. The firm therefore chooses how finely to divide work by trading off the productivity gains from specialization against the coordination costs created by dividing tacit work across workers.

Automation changes this organizational problem through two forces. First, it changes the composition of human work. As automation changes which tasks remain with workers, the human task set can become more tacit. This composition force favors broader jobs because tacit tasks benefit less from specialization and are more costly to coordinate across worker boundaries. Second, automation changes the size of the human task set. When adoption creates more complementary human work than it displaces, the task set expands and the scale force also favors broader jobs. When the human task set contracts, however, the scale force instead favors narrower jobs. Automation therefore has no universal effect on job design. Its organizational consequences depend on how a particular technology changes both what humans do and how much human work remains inside the firm.

I test these predictions using data from Revelio Labs on online job postings and worker employment histories. The posting data cover more than 119 million U.S. vacancies between 2021 and 2025 and identify the employer, occupation, posting date, location, and free-text description of each job. I link these descriptions to O*NET Detailed Work Activities to measure the tasks assigned to individual jobs and use task-level codifiability measures from Schaal (2025) to characterize the tacit content of those tasks. Revelio's employment histories, constructed from LinkedIn profiles, provide complementary information on firms' occupational composition and employment. Together, these data allow me to observe the content of jobs, the composition of firms' human work, and the size of their workforces.

I identify automation adoption using skills demanded in firms' job postings and distinguish three forms of automation: robotics, software, and artificial intelligence. Because firms choose whether and when to adopt these technologies, simple comparisons between adopting and non-adopting firms may confound automation with other changes in firm organization. I therefore construct shift-share instruments that interact firms' predetermined occupational composition with national trends in technology adoption by occupation.

The main result is that different automation technologies reorganize jobs in opposite directions. When treated as a single category, a doubling of automation adoption is estimated to increase the number of tasks assigned to the average job in the firm by approximately 2.7 percent. Disaggregating automation reveals that this average masks heterogeneity: robotics and software broaden jobs while AI significantly narrows them. In the preferred specification, a doubling of robotics adoption increases the number of tasks assigned to the average job in the firm by approximately 27.7%, while a doubling of software adoption increases it by 16.9 %. AI has the opposite effect: a doubling of adoption reduces the number of tasks assigned to the average job by approximately 34.7 %.

I next examine why the organizational response differs so sharply across technologies. The differences track how each technology changes the coordination work that firms assign to workers. Using tasks classified as involving either coordination among workers or between workers and technology, I find that robotics and software increase coordination work in the firm while AI reduces it. This pattern coincides with the headline differences in job breadth: robotics and software increase coordination-related responsibilities and broaden jobs at the firm level, while AI reduces coordination work and narrows jobs. For software, however, the job-breadth effect becomes indistinguishable from zero once occupation is held fixed, suggesting that firm-level broadening from software operates primarily through changes in occupational composition.

Changes in the composition of human work do not follow the same pattern. Robotics and AI both raise the tacit content of the tasks that remain with workers, while software lowers it. Studying the introduction and removal of tasks provides further evidence for the mechanism, though the pattern holds only for robotics and AI. Both technologies remove and also introduce tacit tasks. Software shows evidence of codified-biased task creation. The evidence therefore suggests that another method by which automation can reorganize production is by reallocating the tasks performed by workers.

The paper contributes to the literature on technological change and the organization of work in three ways. First, it shows that automation is not a homogeneous organizational treatment. Technologies that are commonly grouped together as automation can have effects on job design that differ even in sign. Second, it introduces job breadth as an important margin through which firms respond to technological change. The displacement or creation of tasks need not translate directly into changes in employment because firms can also reorganize the remaining work across workers. Third, the results connect the task-based approach to technological change with theories of the division of labor by showing that automation changes the specialization-coordination problem itself. Understanding the consequences of automation therefore requires studying not only which tasks machines perform and how many workers firms employ, but also how firms organize the human work that remains.

My work contributes to two streams of research. The first documents how firms organize work. Smith (1776) first argued that dividing production into specialized tasks raises productivity of workers. Becker and Murphy (1992) build on this work by formalizing a tradeoff within the firm, showing that firms weigh the productivity gains from specialization against the coordination costs of employing a larger workforce. Garicano (2000), on the other hand, models how firms organize knowledge and problem-solving across a hierarchy. He shows that the relative costs of acquiring and communicating knowledge has implications for the allocation of tasks between workers and managers. Relatedly, Holmstrom and Milgrom (1991) show that incentives and monitoring of workers can affect the design of jobs. Together, this literature shows that firms optimally allocate and bundle tasks in response to economic and organizational tradeoffs. The effect of automation on this tradeoff is theoretically ambiguous: while automation changes the tasks performed by workers, it need not directly reduce the coordination costs created by specialization. I contribute to this work by showing how automation technologies can disrupt the traditional tradeoff between specialization and coordination costs.

I also contribute to the literature on the effects of automation technologies. Autor et al. (2003) show that computers substitute for routine tasks while complementing non-routine cognitive work. Building on this idea, Autor (2015) and Autor (2014) argue that automation has historically complemented the tacit, judgment-intensive components of work, reflecting Polanyi's observation that "we know more than we can tell." Acemoglu and Restrepo (2018) and Acemoglu and Restrepo (2019) formalize the relationship between automation and labor, showing that automation can both displace workers from tasks previously performed by labor and create new tasks in which labor retains a comparative advantage. Acemoglu and Restrepo (2020) further provide evidence of the displacement of workers by industrial robotics. I contribute to this literature by documenting how automation has consequences for the intensive margin of job design: automation leads to an expansion in the task scope of workers.

2 Model

2.1 Knowledge and Coordination

A firm produces output using a unit measure of tasks. These tasks range from $[0, 1]$. Tasks require knowledge that ranges from being fully codified to being fully tacit. Knowledge is codified if it can be articulated explicitly. Instructions, formulas, and decision rules are all examples of highly codified knowledge. Tacit knowledge, on the other hand, is knowledge that cannot be fully articulated. This knowledge is often acquired through experience and observation rather than communication. Knowing how a negotiation will unfold, when a dish is ready to be served, and which arguments resonate well with a jury are examples of tacit knowledge. Throughout, I refer to tasks whose knowledge is fully codifiable as codified tasks, and to tasks requiring any amount of knowledge that cannot be articulated as tacit tasks.

Definition 1 (Codifiability)**.** The codifiability of a task i is represented with $c_i \in [0, 1]$, which measures the fraction of task-relevant knowledge that can be explicitly articulated. Each task requires knowledge η_i. When a worker must coordinate tasks pertaining to η_i with their colleagues, the worker can only communicate a representation of it, $\hat{\eta}_i = \eta_i + \varepsilon_i$. The error ε_i is determined by the codifiability of the task:

$$\varepsilon_i^2 \; = \; (1 - c_i)\, \nu^2 \tag{1}$$

where $\nu^2 > 0$ is the squared error on a task with no knowledge that can be explicitly articulated. Tasks with any codified knowledge are thus an improvement, closer representations of η_i, compared to this worst case baseline. A fully codified task has $\varepsilon_i = 0$, so $\hat{\eta}_i = \eta_i$ and no information is lost in communication between workers. All knowledge in this exchange is perfectly articulable. When $c_i < 1$ the transfer of knowledge is incomplete; some information gets lost from one worker to the other and this information loss grows with the tacit content of the task.

Tasks are heterogeneous in codifiability. A fraction $\lambda \in [0, 1]$ of the firm's initial task set is fully codified ($c_i = 1$). The remaining fraction $1 - \lambda$ requires some tacit knowledge and are distributed over $c_i \in [0, c_T]$ according to density $g(c)$, where $c_T < 1$.

A worker knows the true knowledge η_i underlying the tasks assigned to them, but does not know the true knowledge underlying tasks assigned to their colleagues. The loss in (1) therefore arises from the division of labor: when tasks are assigned to different workers, intra-firm coordination requires knowledge to cross worker boundaries, and some of that knowledge is lost whenever $c_i < 1$. I summarize the codifiability of tacit tasks by their average, denoted $\bar{c} \equiv \mathbb{E}[c_i \mid c_i < 1]$.

Three properties follow from codifiability c:

1. **Automatability**: A machine can act only on what has been made explicit. It cannot act on the residual ε_i, so a task is automatable only to the extent that its knowledge has been articulated.

2. **Production loss**: When tacit knowledge must cross a worker boundary, the receiving worker acts on the incomplete representation $\hat{\eta}_i$ rather than on the knowledge itself, and the output of workers is diminished in quality as a result. The loss scales with the tacit share of a task, so codified tasks ($c_i = 1$) generate no loss whatever.

3. **Returns to specialization**: Specialization raises per-task output by allowing workers to devote more time to fewer tasks. The size of this per-task gain depends on knowledge type. For codified tasks, specialization results into large performance improvements. Explicit knowledge can be learned easily through practice, so each additional unit of concentrated time yields a large productivity gain per task. For tacit tasks, expertise develops through exposure to various domains rather than repetition in a narrow area.

Deriving the production loss. A worker acting on $\hat{\eta}_i$ rather than on η_i makes decisions based on an incomplete representation of the knowledge relevant to task i. The worker's error is ε_i, and I assume that the resulting loss in output is the square of this error. Thus, small communication errors have relatively small consequences for output, while larger errors are increasingly costly.

When knowledge about task i must be communicated across a worker boundary, the resulting production loss, denoted by ℓ_i, is therefore

$$\ell_i = \varepsilon_i^2 = (1 - c_i)\nu^2. \tag{2}$$

The production loss from coordinating task i across workers therefore decreases with its codifiability. When $c_i = 1$, the error vanishes and coordination is lossless; when $c_i = 0$, none of the task-relevant knowledge can be articulated and the full loss ν^2 is incurred.

Averaging (2) across the firm's tacit tasks gives the average production loss for each episode of coordination between workers:

$$\ell \;=\; (1 - \bar{c}) \times \nu^2 \;>\; 0, \tag{3}$$

Codified tasks contribute nothing to production losses: their knowledge is written down in full, the error is zero, and the coordination episode between workers is costless.

Scope of information. In addition to the codifiability of the knowledge required to complete a task, the knowledge required to complete a task also varies in its breadth. Some tasks require knowledge over a narrow domain, such as counting currency, whereas other tasks require knowledge over a much broader range of knowledge, such as the knowledge required for project management. I denote this scope by I and measure it as the share of the firm's knowledge domains that a task's requirements span. It varies by knowledge type:

- Codified tasks require narrow information scope, so: I_C is low.
- Tacit tasks require broad information scope, so: I_T is high. Specifically, I assume that $1 \geq I_T > I_C > 0$.

I treat I_C and I_T as fixed characteristics of the firm's tasks. Information scope determines how broadly workers must draw on knowledge from across the firm when performing a task. A task with narrow information scope can more easily be performed using knowledge contained within a worker's own human capital, whereas a task with broad information scope is more likely to require knowledge that spans the knowledge of various workers.

2.2 Specialization of Workers

Firms determine the specialization of their workers by choosing how many workers to employ. Let W denote the number of workers a firm employs. Given a task space of size Ω, each worker is responsible for handling

$$n = \frac{\Omega}{W} \tag{4}$$

tasks. n is the task scope of each worker. Specialization of the worker can then be defined as the inverse:

$$s = \frac{W}{\Omega} = \frac{1}{n} \tag{5}$$

Specialized workers (those with higher s) are responsible for only a narrow set of tasks. When workers are assigned a broader set of tasks, they are considered generalists.

Following Becker and Murphy (1992), I model output as strictly increasing in specialization: a worker assigned to fewer tasks produces more on each one. I extend their framework by allowing the returns to specialization to differ between tacit and codified tasks.

Assumption 1. $1 > \alpha_C > \alpha_T > 0$: the returns to specialization are positive but diminishing for both knowledge types, and codified tasks exhibit higher returns to specialization than tacit tasks.

A worker with fewer tasks produces more per task: per-task output on type-k tasks equals s^{α_k}. Total output for the task space then is:

$$B(s) = \theta_C \cdot s^{\alpha_C} + \theta_T \cdot s^{\alpha_T} \tag{6}$$

where $\theta_C \in [0, 1]$ and $\theta_T = 1 - \theta_C$ denote the codified and tacit shares of the firm's current task set. These are distinct from the fixed parameters λ and $1 - \lambda$ of Section 2.1, which describe the task set only as originally given. At the outset $\theta_C = \lambda$.

$B(s)$ is strictly increasing in s. If not for coordination costs, the firm would want maximum specialization. The marginal benefit is:

$$B'(s) = \alpha_C \, \theta_C \, s^{\alpha_C - 1} + \alpha_T \, \theta_T \, s^{\alpha_T - 1} \tag{7}$$

$B'(s) > 0$ for all $s > 0$.

Since codified tasks generate no coordination loss, coordination costs scale with the tacit share θ_T. Consider a single tacit task and a single worker boundary. That pair generates a costly coordination episode with probability I_T, and if the episode occurs, it costs ℓ. The expected coordination cost of one tacit-task/boundary pair is therefore:

$$\Gamma \equiv \ell \cdot I_T, \tag{8}$$

Γ is the expected cost of one task-boundary pair and is a constant.

Note: I am likely going to remove or change this greyed out section below.

Assumption 2 (Task assignments)**.** The firm's knowledge domain is represented as a circle of circumference one, which the firm partitions into W arcs of equal length $1/W$, one for each worker. Different regions of this space correspond to different areas of knowledge within the firm, such as product, marketing, legal, or information technology. Workers specialize in particular regions of the knowledge space, reflecting the domain-specific human capital they have accumulated through education and experience. The firm assigns each worker those tasks whose knowledge requirements lie primarily within that worker's arc. Each task's required knowledge occupies an interval of length I somewhere on the knowledge domain. The task is performed by the worker whose arc contains the interval's center, and requires coordination with every additional worker into whose arc the interval extends. Figure 2 illustrates the partitioning of knowledge in the firm.

Assumption 3 (Fixed knowledge domain)**.** The size of the knowledge space does not grow with the number of tasks Ω the firm performs.

A firm's knowledge domain is largely determined by the nature of what it produces. I therefore treat changes in the firm's task set as changes in the amount and variety of work performed within an existing knowledge domain, rather than as changes in the breadth of knowledge required by the firm.

A task requires coordination whenever its knowledge requirements extend beyond the worker's own assignment and into another worker's region of the knowledge space. Consider a tacit task with information scope I_T. Since the task occupies an interval of length I_T and the knowledge space is divided by W evenly spaced worker boundaries, the task crosses $I_T W$ boundaries on average. Each crossing represents a coordination episode with another worker.

There are $\theta_T \Omega$ tacit tasks in the firm, so the expected number of costly coordination episodes is $\theta_T \Omega \cdot I_T W$. Recalling that $W = s\Omega$, this becomes $\theta_T I_T s \Omega^2$. Each episode generates an expected production loss ℓ. Using $\Gamma \equiv \ell I_T$ from (8), the firm's total expected coordination cost is therefore

$$C(s; \Omega, \theta_T) = \theta_T s \Omega^2 \Gamma. \tag{9}$$

Equation (9) captures how the division of labor generates coordination costs. Greater specialization raises coordination costs because dividing work among more workers creates more boundaries across which knowledge must be communicated. These costs are larger when the firm's work is more tacit, since only tacit knowledge is lost when it crosses a worker boundary. Finally, coordination costs increase with the square of the task set, Ω. At a given level of specialization, a larger task set implies both more tasks that may require coordination and more workers across whom those tasks are divided. Thus, the benefits of specialization must be weighed against the additional coordination costs created by a finer division of labor.

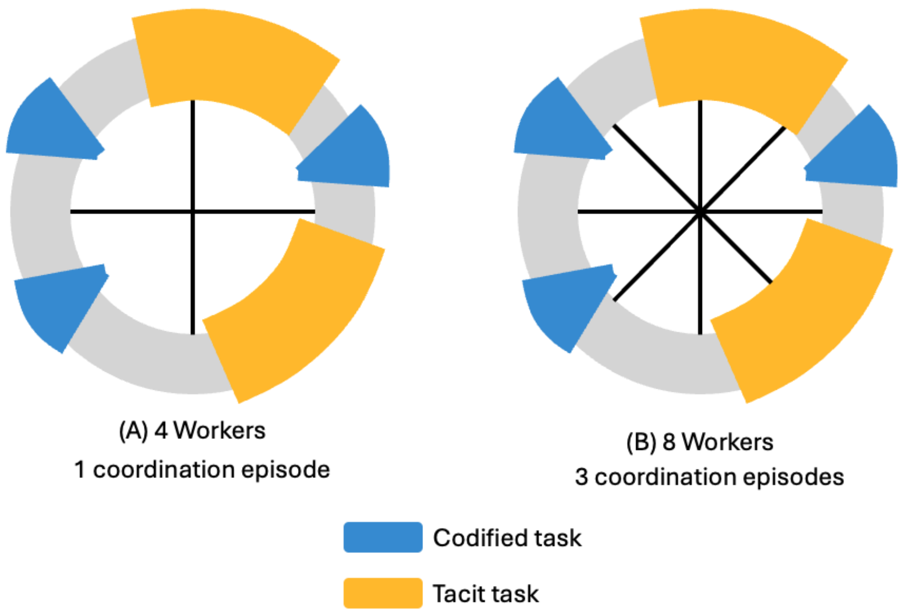

Figure 1: Finer division of labor creates more coordination episodes.
Note: The circles in Panels (A) and (B) of Figure 1 represent the knowledge space for a firm. The blue and orange regions of the space represent tasks that draw from those knowledge intervals for codified and tacit tasks, respectively. Although both firms have the same distribution of tasks and count of tasks, increasing the worker count in panel (B) results in a greater need for coordination. The tacit tasks in panel (B) cross 3 worker boundaries and thus create 3 coordination episodes, while the tacit tasks in panel (A) only generate one coordination episode.

2.3 Automation

Automation technologies perform tasks previously carried out by workers, but their adoption can also create new human tasks associated with monitoring, maintaining, and integrating automated systems. Following Acemoglu and Restrepo (2019), I therefore model automation as both displacing existing human tasks and reinstating new ones. Technologies differ along two dimensions: the kinds of tasks they can perform and the amount of complementary work their adoption requires. I index these differences by technology type τ.

Definition 1 characterizes tasks by the extent to which their required knowledge can be articulated. Automation technologies differ in how much codifiability they require in order to perform a task. I capture this difference by the technology's reach, $\bar{c}_\tau \in [0,1]$. A task i is within the reach of technology τ if

$$c_i \geq \bar{c}_\tau. \tag{10}$$

A technology with $\bar{c}_\tau = 1$ can perform only fully codified tasks. This describes technologies that execute procedures specified in advance and therefore require the relevant knowledge to be fully articulated. Technologies that infer patterns from data may require less explicit specification and therefore have $\bar{c}_\tau < 1$, allowing them to reach tasks containing some tacit knowledge. In this sense, a decline in $\bar{c}_\tau$ captures the erosion of Polanyi's paradox described by Autor (2014). Figure 2 demonstrates how different tasks are eligible for automation given a technology's reach.

Reach is a property of the technology rather than of the task. The same task may therefore lie within the reach of one technology but outside the reach of another. The case of $\bar{c}_\tau = 1$ can serve as a benchmark in which automation can displace fully codified tasks but cannot displace tasks containing tacit knowledge.

Reach determines which tasks are eligible for automation, but does not by itself determine which eligible tasks are displaced. I therefore impose the following restriction on selection within the eligible set.

Assumption 4 (Unselective displacement)**.** Conditional on being within the technology's reach, tasks are equally likely to be displaced.

The assumption separates technological capability from task selection. Codifiability determines whether technology τ can perform a task; conditional on satisfying that constraint, displacement is not further selected according to codifiability.

Recall that tacit tasks have codifiability distributed over $c \in [0, c_T]$ with density $g(c)$.

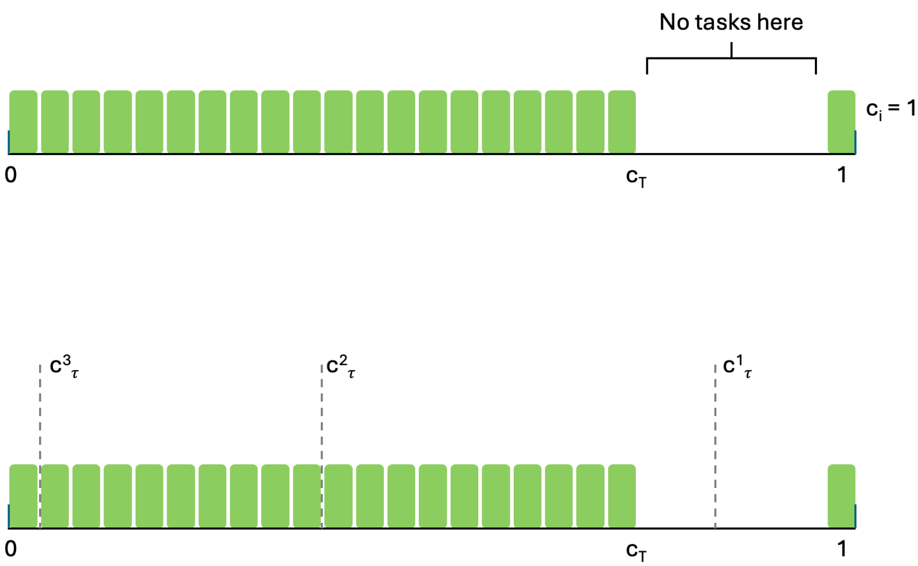

Figure 2: Demonstration of relationship between codifiability and automatability.
Note: Figure 2 represents the relationship between codifiability and automatability. The continuum shows that the firm has some perfectly codified tasks at $c_i = 1$, and has tacit tasks ranging from $c_i = 0$ through c_T. The firm uses no tasks ranging from c_T to $c_i = 1$. A technology with reach c_τ^1 would only have the capabilities to automate perfectly codified tasks. A technology with reach c_τ^2 would be capable of automating some tacit tasks, those with $c_i > c_\tau^2$, as well as perfectly codified tasks, whereas a technology with reach c_τ^3 would be capable of automating all but the most tacit tasks in the firm. Among automatable tasks, the firm is indiscrimant amongst which it automates.

For a technology with reach $\bar{c}_\tau$, the mass of tacit tasks eligible for automation is

$$\mu_\tau \equiv \int_{\bar{c}_\tau}^{c_T} g(c)\,dc, \tag{11}$$

where $\mu_\tau = 0$ when $\bar{c}_\tau \geq c_T$. Because fully codified tasks have $c_i = 1$, the entire codified mass λ is eligible for every technology. The total mass of eligible tasks is therefore $\lambda + \mu_\tau$.

Under Assumption 4, displaced tasks reflect the composition of this eligible set. The tacit share of displaced work is therefore

$$\delta_\tau \equiv \frac{\mu_\tau}{\lambda + \mu_\tau}. \tag{12}$$

Technology reach thus determines the composition of displaced work. When $\bar{c}_\tau \geq c_T$, no tacit tasks are within reach and $\delta_\tau = 0$. As $\bar{c}_\tau$ falls, more tacit tasks become eligible and δ_τ rises. At $\bar{c}_\tau = 0$, all tasks are eligible, so the displaced task mix reproduces the firm's initial task composition and $\delta_\tau = 1 - \lambda$.

Adopting an automation technology may also require organizational changes and new human tasks. Automated systems must be installed, calibrated, monitored, repaired, and integrated with the firm's existing production process (Bresnahan et al., 2002; David, 1990; Brynjolfsson and Hitt, 2000; Bartel et al., 2007). Technologies differ in the amount of such complementary work they require.

I capture this difference with the technology's task reinstatement rate, $\chi_\tau \geq 0$, defined as the number of new human tasks introduced within the firm per task displaced by automation. Thus, displacing one task creates χ_τ new tasks that must continue to be performed by workers.

I assume that these reinstated tasks are fully tacit, with $c_i = 0$. The idea is that the complementary work remaining inside the firm is highly contextual and specific to the firm's production environment: work that could instead be fully specified can more readily be incorporated into the technology itself. This assumption also implies that reinstated tasks cannot themselves be automated by any technology with positive reach, $\bar{c}_\tau > 0$.

Let A denote adoption intensity, measured as the mass of existing tasks displaced by technology τ. Adoption therefore removes A tasks and introduces $\chi_\tau A$ complementary tasks. Beginning from a unit task space, the total mass of human tasks after automation is

$$\Omega_\tau(A) = 1 + (\chi_\tau - 1)A. \tag{13}$$

Because a fraction δ_τ of displaced tasks are tacit, the post-automation masses of tacit and codified work are

$$T_\tau(A) = (1-\lambda) - \delta_\tau A + \chi_\tau A, \qquad K_\tau(A) = \lambda - (1-\delta_\tau)A. \tag{14}$$

The corresponding tacit share of the human task set is

$$\theta_T(A) \equiv \frac{T_\tau(A)}{\Omega_\tau(A)}. \tag{15}$$

These expressions separate two effects of automation. The displacement composition δ_τ determines which kinds of existing tasks are removed, while the task reinstatement rate χ_τ determines how the size of the human task set changes. If $\chi_\tau > 1$, adoption creates more human tasks than it displaces and the task set expands. If $\chi_\tau < 1$, the task set contracts. When $\chi_\tau = 1$, each displaced task is replaced by one complementary task, so $\Omega_\tau(A) = 1$ and automation changes the composition of human work without changing its total size.

Differentiating the tacit share with respect to adoption gives

$$\frac{d\theta_T}{dA} = \frac{(\chi_\tau - \delta_\tau) - \theta_T(\chi_\tau - 1)}{\Omega_\tau(A)}. \tag{16}$$

Equation (16) captures the net change in the composition of human work. Automation removes tacit work at rate δ_τ, adds tacit complementary work at rate χ_τ, and may simultaneously change the size of the task set. In the case where $\chi_\tau = 1$, task-set size is fixed and $d\theta_T/dA = 1 - \delta_\tau \geq 0$; automation shifts the remaining human task set toward tacit work whenever it displaces any codified tasks.

2.4 The Firm's Problem

The firm chooses how finely to divide work across workers. Given automation intensity A, it chooses the specialization level s to maximize profits. Gross output increases with specialization, while greater specialization also creates more worker boundaries across which tacit knowledge must be coordinated. The firm's problem is therefore

$$\Pi(s; A) = \Omega_\tau(A)\left[\theta_C(A)s^{\alpha_C} + \theta_T(A)s^{\alpha_T}\right] - \theta_T(A)\Gamma s \Omega_\tau(A)^2. \tag{17}$$

The first term is gross output. The expression in brackets is output per unit of the task space, $B(s)$, and $\Omega_\tau(A)$ scales this output by the total mass of human tasks. The second term is the coordination cost derived in equation (9). Thus, specialization raises output by allowing workers to focus on narrower sets of tasks, but it also raises coordination costs by increasing the number of worker boundaries.

The optimal specialization level s^* satisfies

$$B'(s^*; A) = \theta_T(A)\Gamma\Omega_\tau(A). \tag{18}$$

Equation (18) summarizes the firm's organizational trade-off. The left-hand side is the marginal benefit of greater specialization, while the right-hand side is its marginal coordination cost. This cost is larger when a greater share of human work is tacit, when knowledge loss across worker boundaries is greater, or when the firm performs a larger set of tasks.

In order to simplify the comparative statics below, I evaluate the first-order condition at the normalization $s^* = 1$.

2.5 Predictions

Automation changes the firm's optimal division of labor through two characteristics of the human task set: its composition and its size. Changes in composition affect both the productivity benefits of specialization and the costs of coordinating work across workers. Changes in the size of the task set affect how many worker boundaries are required to sustain a given degree of specialization.

When automation leaves the firm's human work more tacit, two changes occur. First, the average return to specialization falls because tacit tasks benefit less from narrow specialization. Second, a larger share of the firm's work becomes subject to information loss when tasks are divided across workers. Since both effects arise from the changing composition of work, I refer to their combined effect as the composition force.

Changes in the size of the human task set operate through a different mechanism. When the firm performs more distinct tasks, maintaining a given degree of specialization requires more workers and therefore more boundaries across which tacit knowledge must be coordinated. I refer to this effect as the scale force.

The effect of automation on job design can therefore be summarized by these two forces.

Proposition 1 (Automation affects the size and composition of tasks)**.** *Define*

$$\Psi \equiv \frac{\alpha_C - \alpha_T}{\bar{\alpha}} + \frac{1}{\theta_T} = \frac{\alpha_C}{\bar{\alpha}\theta_T} > 0. \tag{19}$$

Then the response of optimal specialization to automation is

$$\frac{d \ln s^*}{dA} = -\frac{\bar{\alpha}}{\kappa}\left[\Psi \frac{d\theta_T}{dA} + \frac{\chi_\tau - 1}{\Omega_\tau(A)}\right], \qquad \kappa \equiv -B''(s^*) > 0. \tag{20}$$

The first term inside the brackets represents a recomposition of the task set and the second represents a change in the size of the task set.

Equation (20) shows how the direction of each force depends on the characteristics of the adopted technology. The first term in the bracket pushes toward broader jobs whenever automation increases the tacit share of remaining work, $d\theta_T/dA > 0$. The second termn pushes toward broader jobs when the technology expands the human task set, $\chi_\tau > 1$, and toward narrower jobs when it contracts the task set, $\chi_\tau < 1$. Whether automation ultimately broadens or narrows jobs therefore depends on the direction and relative strength of these two forces.

Proposition 2 (Composition of remaining work)**.** *For every technology type and every adoption intensity A,*

$$\frac{d\theta_T}{dA} \geq 0. \tag{21}$$

Automation always weakly increases the tacit share of the remaining work.

The result follows from the composition of the technology's eligible task set. Fully codified tasks are within the reach of every technology, while access to tacit tasks depends on $\bar{c}_\tau$. Under Assumption 4, the tacit share of displaced work can therefore never exceed the tacit share of the firm's task set. Automation may leave the composition of human work unchanged or shift it toward more tacit work, but under Assumption 4 it cannot shift the remaining human task set toward more codified work.

Technology reach affects the strength of this compositional change. As $\bar{c}_\tau$ falls, more tacit tasks become eligible for automation and δ_τ rises. The work displaced by the technology consequently becomes more similar in composition to the firm's existing task set, weakening the change in the composition of the work that remains. Greater reach can therefore weaken the impacts of recomposition, but it cannot reverse its direction.

The model thus generates that organizations may respond in two ways to automation. Recompostion should cause the firm to broaden jobs, while resizing of the task set should broaden jobs if the task set expands or narrow jobs if the task set contracts. In the next sections, I will perform tests of these predictions.

3 Data

In this section, I describe the data on job postings, employment histories, and tasks used to construct the empirical measures of job breadth and automation adoption.

3.1 Job Postings and Employment Histories

Job postings provide information on the work that firms assign to individual workers. Revelio Labs collects and deduplicates online job postings from company career pages and third-party job boards. Each posting contains the employer, job title, posting date, location, and free-text description of the position. The free-text description is particularly useful for this paper because it contains information that describes the responsibilities expected of the hired worker. Revelio also provides employment histories constructed from LinkedIn profiles. These records identify workers' employers, occupations, and employment dates. Together, the job postings and employment data allow me to observe both the content of firms' jobs and the composition and size of their workforces.

3.2 Occupations and Tasks

O*NET is a publicly available data sources maintained by the U.S. Department of Labor which provides a standardized taxonomy for organizing occupations and tasks. O*NET classifies 1,016 occupations using Standard Occupational Classification codes. Revelio maps the job title on job postings to these codes. O*NET also maintains a taxonomy of tasks. At the coarsest level, ONET delinates 41 Generalized Work Activities (GWAs). These represent the broadest classification of tasks, like "Handling and Moving Objects" and "Performing Administrative Activites". GWAs are further divided into 332 Intermediate Work Activities (IWAs). This finer classification, including entries such as "Stock supplies or products" and "Perform financial transactions" represent a narrower definition of work. Finally, Detailed Work Activities (DWAs) represent an even more granular classification of work. The 2,087 DWAs include entries such as "Stock medical or patient care supplies" and "Operate cash registers". DWAs will be used to represent tasks in the empirical analysis.

3.3 Task Codifiability

The task-level codifiability scores from Schaal (2025) provide a measure of the knowledge required to perform each O*NET DWA. Three large language models were used in Schaal (2025) to evaluate the tacit knowledge required by each task. The resulting composite score ranges from zero to two, with higher values indicating greater codifiability and lower values indicating greater tacitness. The scores capture differences in the knowledge required across tasks. Activities such as counting money, maintaining records of goods received, and recording operational information receive high codifiability scores. Activities such as negotiating contracts, presiding over a board of directors, and reviewing and analyzing legislation receive low scores. These scores allow the empirical analysis to relate the task content observed in job postings to the distinction between tacit and codified knowledge developed in the model.

3.4 Sample

I restrict the sample to U.S. firms observed in the job-posting data between 2021 and 2025. To reduce measurement error from sparse posting activity, I retain only firm-year observations with at least 10 job postings spanning at least three distinct occupations. These restrictions make the observed task content more likely to reflect a representative portion of a firm's hiring activity rather than a small number of isolated vacancies.

The sample is limited in how representative of the population of U.S. firms it is. Firms with relatively little online hiring activity are less likely to enter the sample, so smaller firms, firms with a limited digital presence, and firms that rely less heavily on online recruiting are likely to be underrepresented. The estimates should therefore be interpreted as applying primarily to firms with sufficiently active and diverse online hiring activity rather than to the full population of U.S. firms.

After applying these restrictions, the sample contains 194,240 firms, 586,487 firm-year observations, and 119,420,460 job postings. The number of observations used in individual analyses varies with the availability of the data required for each outcome. I report the corresponding sample size with each empirical specification.

3.5 Characteristics of Job Postings and Firms

Panel A of Table 1 demonstrates characteristics of job postings in the sample. The average job posting if of modest length, containing 494.7 words. Job postings vary greatly in their content, with the sample consisting of many short postings with few tasks, regardless of the task measure. Job posting content is highly right skewed, with the most length job postings containing nearly 17,000 words, roughly the length of an academic article.

Panel B demonstrates the characteristics of firms in the sample. Reported LinkedIn employment, posting volume, and the number of occupations in which firms hire are right skewed, with a relatively small number of firms accounting for particularly high levels of employment and online hiring activity. The firms in the sample are also older and larger on average than firms in the broader U.S. economy. This reflects the sample's focus on employers with sustained online hiring activity. The results should therefore be interpreted as most representative of established, actively hiring firms rather than the broader population of U.S. firms.

Table 1: Descriptive Statistics

Variable	N	Mean	SD	Min	P_{25}	P_{50}	P_{75}	Max
A. Job posting characteristics								
Word count per posting	119,420,460	494.7	313.6	10.000	279.0	431.0	641.0	16,976
Tasks per posting	119,420,460	23.579	21.015	1.000	7.000	16.000	35.000	70.000
IWAs per posting	119,420,460	14.515	11.552	1.000	5.000	11.000	21.000	57.000
GWAs per posting	119,420,460	9.286	5.909	1.000	4.000	8.000	13.000	30.000
B. Firm characteristics								
LinkedIn Employment	588,000	323.45	4,418.96	1.000	12.500	33.250	93.000	878,014.00
Firm age	598,899	31.849	33.42	1.000	10.500	21.500	41.000	397.00
Postings per firm-year	598,899	199.40	1,857.50	3.000	11.000	24.000	68.000	290,695
6-digit occupations per firm-year	598,899	16.790	28.546	1.000	5.000	8.000	17.000	935.00
Years firm appears in data	194,240	4.314	0.882	2.000	4.000	5.000	5.000	5.000

4 Effects of Automation on Job Breadth

The following section explores the effects of automation on job breadth.

4.1 Empirical Strategy

A simple regression of job breadth on automation adoption is unlikely to recover a causal effect because both are choices made by the firm. Firms that are growing, better managed, or more technologically intensive may be more likely to adopt automation and may organize work differently, even in the absence of automation adoption. To address this concern, I use a shift-share instrument that predicts firms' automation adoption.

The baseline specification I use takes on the following form. For each automation technology τ, I estimate the following first stage:

$$a^{\tau}_{ft} = \pi_{\tau} Z^{\tau}_{ft} + X'_{ft}\rho_{\tau} + \lambda_{j(f),t} + u^{\tau}_{ft}, \tag{22}$$

and second stage:

$$\log(\text{Breadth}_{ft}) = \beta_{\tau}\widehat{a}^{\tau}_{ft} + X'_{ft}\gamma_{\tau} + \lambda_{j(f),t} + \varepsilon^{\tau}_{ft}. \tag{23}$$

Here, Breadth_{ft} is the average number of tasks assigned to a job posted by firm f in year t, and a^{τ}_{ft} measures the firm's accumulated adoption of automation technology τ through the preceding year. There are three types of automation technologies: robotics, software, and artifical intelligence (AI). I also pool these technologies together to examine the effects of automation adoption, in aggregate, on job breadth. The instrument Z^{τ}_{ft} predicts this adoption using the firm's baseline occupational composition, the job composition of the firm in the first year it appears in the data, and national occupation-level adoption of the same technology. The vector X_{ft} contains predetermined and lagged firm characteristics, while $\lambda_{j(f),t}$ denotes four-digit NAICS industry-by-year fixed effects. Standard errors are clustered at the firm level.

The coefficient of interest, β_τ, measures how increased adoption of technology τ changes job breadth. A positive coefficient indicates that automation broadens jobs by increasing the number of tasks assigned to the average role, while a negative coefficient indicates that automation narrows jobs.

4.2 Construction of the Main Variables

Equations (22) and (23) require measures of job scope, automation adoption, and firms' predicted exposure to automation. I construct the first two from the task and skill content of firms' job postings and the third from firms' baseline occupational composition.

4.2.1 Job Breadth

Job breadth refers to the average number of tasks that appear in job postings in a particular firm-year. Job breadth serves as the empirical counterpart to the inverse of specialization in the model: broader jobs contain more tasks per worker and therefore correspond to less specialization, while narrower jobs contain fewer tasks and correspond to greater specialization.

For each firm-year, I average the number of assigned tasks across all of the firm's postings:

$$\mathrm{Scope}_{ft} = \frac{1}{N_{ft}} \sum_{p \in \mathcal{P}_{ft}} |\mathcal{D}_p| \,, \tag{24}$$

where $\mathcal{P}_{ft}$ is the set of postings by firm f in year t, N_{ft} is the number of those postings, and $\mathcal{D}_p$ is the set of unique DWAs assigned to posting p. Higher values therefore correspond to jobs that bundle together a broader set of tasks and thus to less specialization.

I use a sentence embedding model to assign tasks to job postings. Job postings that bear a high textual similarity to a particular task label are assigned to that task. Any postings that returned no match to a task were dropped from the sample. I manually inspected a sample of tasks to ensure the matching algorithm was assigning tasks appropriately.

The baseline measure of tasks uses DWAs because they provide the most detailed standardized description of work in O*NET. I also reconstruct job breadth using broader levels of the O*NET task hierarchy, such as IWAs and GWAs, and obtain similar results. These alternative definitions are reported in the Appendix.

4.2.2 Automation Adoption

I measure automation adoption from the technology-related skills demanded in firms' job postings. To an extent, firms reveal their technological investments through their job postings, because they must hire workers with the skills necessary to operate their machines. For example, a firm seeking workers that can perform robotic programming likely has invested in robotics. The sudden occurrence of a skill related to an automation technology might indicate that the firm has, soon plans to, adopt that technology. A shortcoming of this method is that it cannot distinguish between firms that have adopted automation technologies and firms that sell automation products or services. However, I attempt to mitigate this concern in my construction of the automation measure.

Revelio maintains a standardized list of more than 500 skills that appear across all job postings. Of these, I classified 57 to be related either to robotics, software automation, or artificial intelligence and machine learning (documented in Table 9 in the Appendix). Given that some skills have synonyms, I expand this list to include alternative phrases that represent the same skills. I document 59 phrases pertaining to robotics, skills, 60 pertaining to artificial intelligence or machine learning, and 153 pertaining to software automation. The three categories capture forms of automation that differ in the kinds of work they perform. Robotics includes technologies used to automate physical production activities. Software automation include technologies that automate digital processes. AI includes technologies that use machine learning to perform tasks which don't rely on hardcoded rules.

For each technology τ, I construct a cumulative stock of the distinct automation-related skills that firm f has demanded through year t. A particular skill enters the stock once, in the first year in which the firm's postings demand it; repeated appearances of the same skill do not increase the stock. Letting A^{τ}_{ft} denote this cumulative count, the treatment entering Equation (23) is

$$a^{\tau}_{ft} = \log\left(1 + A^{\tau}_{f,t-1}\right). \tag{25}$$

I use a lagged version of this stock variable by one year so that job breadth in year t is related to automation capabilities accumulated through year $t-1$.

4.2.3 Shift-Share Instrument

I construct a technology-specific shift-share instrument to instrument the endogenous automation adoption variable. Shift-share instruments are commonly used to exploit predetermined differences in units' exposure to a common external source of variation (Bartik, 1991; Autor et al., 2013; Acemoglu and Restrepo, 2020; Forman et al., 2012). The intuition underlying the instrument's construction is as follows: firms differ in the occupations in which they hire and occupations differ in the rate at which particular automation technologies diffuse nationally. Occupations that demonstrate increased signals of automation adoption might reflect that the occupation is undergoing a common supply shock, such as a drop in prices. The adoption of automation occurs at different rateas across occupations, with some occupations demonstrating the signs of high automation adoption, whereas others may show no signs at all. A firm that is composed of many occupations experiencing adoption of a technology nationwide may therefore be more likey to adopt automation themselves. The identifying assumption for the instrument is that, conditional on observables, predicted exposure to adoption trends affects job design only through the firm's own automation adoption and not through alternative channels.

Let $\omega_{fo,0}$ denote occupation o's share of firm f's postings in a baseline period for the firm. For each technology τ, occupation o, and year t, I calculate a national adoption trend $G^{\tau}_{-f,-s(f),ot}$. Here, $-f$ indicates that the focal firm is excluded, $-s(f)$ indicates that all other firms located in the focal firm's state are excluded, o indexes occupation, t indexes year, and τ indexes the automation technology. The technology-specific instrument is

$$Z^{\tau}_{ft} = \sum_{o} \omega_{fo,0} G^{\tau}_{-f,-s(f),ot}. \tag{26}$$

The leave-out construction prevents the focal firm's own adoption from mechanically contributing to its instrument, and also removes contemporaneous technology adoption by other firms in its state. Variation in Z^{τ}_{ft} therefore comes from the interaction between differences in firms' predetermined occupational composition and national technology trends occurring outside their firm and state.

The source of identifying variation also determines how the estimates should be interpreted. Because firms differ in their exposure through occupational shares fixed at baseline, the shift-share design relies primarily on differences across firms rather than solely on changes within the same firm over time. The baseline estimates therefore compare firms in the same industry and year that entered the sample with different exposure to subsequent occupation-specific automation trends.

4.2.4 Controls and Fixed Effects

The baseline specification controls for predetermined differences across firms that may predict both automation adoption and job design. The vector X_{ft} contains the firm's average posting word count in a baseline year, lagged log employment, lagged firm age. Baseline posting length accounts for persistent differences in how extensively firms describe jobs, which could otherwise mechanically affect measured job breadth. The employment and age controls account for differences in firm scale. I also employ an industry-by-year fixed effect, which absorbs common shocks to all firms in the same industry and year. I define an industry as a NAICS 4-digit industry group, of which all 308 industry groups are represented in the sample. The use of this industry-by-year fixed effect means that my identification relies primarily on differences across firms within the same industry and year, rather than changes within the same firm over time.

5 Effects of Automation on Job Breadth

The model's primary prediction is that automation changes the degree of worker specialization: greater specialization should occur if there is substantial displacement of labor from existing tasks, whereas reduced specialization should be observed if labor is largely reinstated into new work.

Table 2 reports the aggregate estimates in columns (1) and (2). Column (1) presents the OLS estimate of equation (23); column (2) presents the corresponding 2SLS estimate, instrumenting for treatment with the shift-share instrument.

The OLS specification shows a strong positive relationship between aggregate automation adoption and job breadth. The OLS coefficient is 0.0630, indicating that firms with greater automation adoption tend to post broader jobs, even after conditioning on the controls and industry-by-year fixed effects. Once aggregate automation adoption is instrumented, the coefficient falls to 0.0390 in column (2). This attenuation is consistent with positive selection into automation in the OLS specification: firms that are growing or undergoing broader organizational change may both adopt more automation and design broader jobs. The instrument isolates variation in automation adoption that is plausibly unrelated to these firm-specific determinants of job design. To the extent that these omitted factors would bias the OLS estimate upward, removing this source of endogeneity would lead to a smaller 2SLS estimate. Taken together, the estimates indicate that automation adoption, in aggregate, increases job breadth.

Column (3) then disaggregates automation into the three technology types – robotics, software, and AI – entered jointly as endogenous regressors and instrumented simultaneously by their corresponding shift-share instruments. Rather than reproducing the positive aggregate effect, the three technologies move job breadth in different directions. The coefficient on robotics is 0.3529, implying that a doubling of the robotics stock increases the number of tasks assigned to the average job by approximately 27.7 percent. Software produces a similar broadening effect: the coefficient is 0.2245, implying that a doubling of the software stock

Table 2: Effects of Automation on Job Breadth

	Outcome: log(Breadth)			
	Aggregate Automation		Decomposed	
	OLS	2SLS	2SLS	2SLS
	(1)	(2)	(3)	(4)
log(Automation)	0.0630***	0.0390***		
	(0.0017)	(0.0043)		
log(Robotics)			0.3529***	0.0892***
			(0.0114)	(0.0086)
log(Software)			0.2245**	0.0664
			(0.1032)	(0.0586)
log(AI)			-0.6149***	-0.2140***
			(0.0575)	(0.0434)
First-stage F	–	7390.9		
First-stage F: Robotics			3149.1	2172.5
SW partial F: Robotics			3079.7	1928.1
First-stage F: Software			2084.2	994.6
SW partial F: Software			130.8	49.5
First-stage F: AI			1510.2	728.2
SW partial F: AI			111.7	37.5
Observations	586,487	586,487	563,115	9,067,989
Firms	191,568	191,568	191,071	191,073
Controls	Yes	Yes	Yes	Yes
NAICS-4 × Year FE	Yes	Yes	Yes	Yes
Occupation × Year FE	No	No	No	Yes

Notes: Column (4) is estimated at the occupation x firm x year level, with occupation-by-year fixed effects added. SW partial F reports the Sanderson-Windmeijer partial first-stage F-statistic for each endogenous regressor net of the identifying content already spanned by the other two instruments. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

increases job breadth by approximately 16.8 percent. AI, by contrast, has the opposite effect: the coefficient of -0.6149 implies that a doubling of the AI stock reduces job breadth by approximately 34.7 percent. The positive aggregate effect in column (2) therefore conceals substantial heterogeneity across automation technologies.

The firm-year estimates may reflect changes in the occupational composition of firm's hiring rather than changes in the design of jobs. To separate these margins, I re-estimate the specification at the occupation-firm-year level and include occupation-by-year fixed effects. This specification compares firms hiring into the same occupation in the same year, thereby isolating differences in job breadth within occupations while absorbing common occupation-specific shocks. The results are reported in column (4). The robotics and AI coefficients retain their signs but become smaller in magnitude, suggesting that occupational reallocation accounts for part of their firm-level effects on job breadth. The software coefficient, by contrast, becomes indistinguishable from zero. This suggests that the firm-level broadening associated with software adoption operates primarily through changes in the firm's occupational composition rather than through broader task assignments within a given occupation.

DWAs are my preferred unit for measuring specialization because they represent the finest level of work activity in the O*NET hierarchy. As a robustness check, I reconstruct job breadth using the coarser IWA and GWA levels. Whereas changes in DWA breadth primarily capture changes in the number of specific activities assigned to a job, changes at these coarser levels capture whether workers are assigned responsibilities spanning a wider range of work. They therefore provide an alternative measure of specialization that places greater weight on the variety of activities bundled within a job. Appendix B reports these results. The technology-specific pattern is unchanged: robotics and software broaden jobs, while AI narrows them.

The effects reported in Table 2 may suggest that different forms of automation may alter firms' coordination requirements in different ways. Robotics and software are more rigid technologies whose adoption may generate complementary activities such as technology implementation, integration, monitoring, and maintenance. AI, however, may perform or facilitate some activities that would otherwise require coordination across workers. Differences in job specialization may therefore reflect not only which existing tasks a technology automates, but also the coordination work that its adoption creates or absorbs. This interpretation is related to Garicano et al. (2026), who show theoretically that whether technology narrows an occupation's task boundary depends on the cost of separating the tasks performed by the technology from the remainder of the job. In this setting, unbundling costs may depend on whether a technology substitutes for coordination or instead creates new coordination requirements. AI may perform or facilitate activities that previously coordinated work across employees, reducing coordination costs and permitting a finer division of labor. Robotics and software may instead require workers to coordinate not only with one another but also with the newly introduced technology, creating complementary implementation, monitoring, maintenance, and troubleshooting responsibilities. These additional human–machine interfaces can increase the coordination burden of production and favor broader task bundles.

6 Effects of Automation on Coordination Work

I next examine whether differences in coordination requirements can help explain why automation technologies have different effects on specialization. AI may substitute for coordination-intensive activities, reducing the coordination burden borne by workers and permitting a finer division of labor. Robotics and software may instead create additional coordination requirements as workers integrate, monitor, and interact with the technology, favoring broader task bundles. I distinguish between two types of coordination work: coordination between workers, and coordination between a worker and a machine.

I examine both worker–worker coordination and worker–technology coordination using the following specification. For each automation technology τ, the first stage is the same as Equation (22), with occupation-by-year fixed effects $\mu_{o,t}$ added alongside $\lambda_{j(f),t}$. I estimate the following second stage

$$Y^{g}_{fot} = \sum_{\tau} \beta^{g}_{\tau}\widehat{a}^{\tau}_{ft} + X'_{ft}\gamma^{g} + \lambda_{j(f),t} + \mu_{o,t} + \varepsilon^{g}_{fot}, \tag{27}$$

estimated at the firm-by-occupation-by-year level. Here, Y^{g}_{fot} is the log of one plus the average number of coordination tasks per job posting by firm f in occupation o in year t, for $g \in \{1, 2\}$, where $g = 1$ denotes worker–worker coordination and $g = 2$ denotes worker–technology coordination. As in Equation (23), a^{τ}_{ft} is the firm's accumulated adoption of automation technology τ through the preceding year and Z^{τ}_{ft} is the corresponding occupation-based shift-share instrument; X_{ft} contains the same predetermined and lagged firm characteristics and $\lambda_{j(f),t}$ again denotes four-digit NAICS industry-by-year fixed effects. The three technology stocks enter Equation (27) jointly and are instrumented simultaneously by their corresponding technology-specific instruments. The new term $\mu_{o,t}$ denotes occupation-by-year fixed effects, so that β^{g}_{τ} is identified from variation across firms posting into the same occupation in the same year, rather than from variation in which occupations firms hire into.

I construct Y^{g}_{fot} directly from the tasks assigned to each job posting. Let $\mathcal{D}_p$ denote the set of tasks assigned to posting p, let G_g denote the set of tasks comprising coordination group g, and let $\mathcal{P}_{fot}$ denote the set of postings by firm f in occupation o in year t. The number of group-g coordination DWAs appearing in posting p is

$$n_p^g = \sum_{d \in \mathcal{D}_p} \mathbb{1}\left[d \in G_g\right], \tag{28}$$

a simple count of the tasks in posting p that fall into group g. Averaging this count across all postings in a firm-occupation-year cell and taking logs gives the outcome used in equation (27):

$$Y_{fot}^g = \log\left(1 + \frac{1}{|\mathcal{P}_{fot}|} \sum_{p \in \mathcal{P}_{fot}} n_p^g\right). \tag{29}$$

To measure coordination work, I manually classify O*NET DWAs according to whether they involve coordination among workers or interaction between workers and technology. I identify 21 DWAs as involving interpersonal coordination and 23 as involving human–machine interaction. Appendix C.5 reports the complete set of DWAs in each category.

Table 3 presents the results of running Equation (27) on each set of coordination tasks. Running this analysis at the occupation x firm x year level allows me to hold occupation fixed when comparing firms, so the estimated effect reflects how automation changes the coordination work rather than which occupations a firm chooses to hire into. Without this, a firm's automation adoption could appear to change coordination or interface task content simply because it shifted its hiring toward different occupations, not because any individual job's coordination requirements actually changed.

Column (1) reports the effects on interpersonal coordination tasks. Robotics and software increase the amount of coordination work assigned to workers, whereas AI reduces it. Column (2) reveals the same pattern for human–machine interaction: robotics and software increase tasks involving interaction with technology, while AI reduces them. These effects closely mirror the technology-specific changes in job breadth: the technologies that broaden jobs also increase coordination-related responsibilities, while AI both narrows jobs and reduces them. The alignment between these outcomes provides evidence consistent with the proposed coordination mechanism: some technologies generate new coordination demands, while others reduce the coordination required of workers. These results hold across both forms of coordination considered, whether among workers or between workers and machines.

A potential concern is that the coordination results are mechanically related to the headline job-breadth estimates because coordination and interface tasks are themselves included in the task count used to construct job breadth. Thus, the results in Table 3 might reflect a mechanical overlap with the outcome rather than evidence of the coordination channel.

Table 3: Effects of Automation on Coordination Work

	log(Interpersonal coordination) (1)	log(Machine coordination) (2)
log(Robotics)	0.0095***	0.0564***
	(0.0025)	(0.0027)
log(Software)	0.1699***	0.1172***
	(0.0188)	(0.0150)
log(AI)	-0.0948***	-0.1050***
	(0.0139)	(0.0108)
First-stage F: Robotics	2121.3	2121.3
SW partial F: Robotics	1883.6	1883.6
First-stage F: Software	965.1	965.1
SW partial F: Software	46.3	46.3
First-stage F: AI	692.8	692.8
SW partial F: AI	35.0	35.0
Observations	8,488,406	8,488,406
Firms	191,064	191,064
Controls	Yes	Yes
NAICS-4 $\times$ Year FE	Yes	Yes
Occupation $\times$ Year FE	Yes	Yes

Notes: The dependent variable in each column is the log of one plus the average count of that integrative-work group's DWAs per job posting, measured at the occupation x firm x year level. Both columns include the same controls as Table 2 (baseline average posting word count, lagged log employment, and lagged firm age). All specifications include four-digit NAICS-by-year and occupation-by-year fixed effects, so identification comes from firms posting into the same occupation in the same year. Standard errors are clustered at the firm level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

To rule out this circularity, I reconstruct the headline outcome after excluding interpersonal coordination tasks, human–machine interface tasks, and both sets of tasks together, and then re-estimate the baseline specifications. The resulting coefficients are nearly unchanged in both sign and magnitude: across all three exclusions, with the largest change departing from the baseline estimate by 10.6 percent. Thus, the changes in specialization are not mechanically driven by the addition or removal of coordination tasks themselves. Rather, changes in coordination requirements may induce firms to reorganize a broader set of tasks within jobs. For example, if AI substitutes for coordination activities, it may reduce the coordination burden faced by workers and allow firms to divide the remaining work more finely. Robotics and software may instead increase coordination requirements, encouraging firms to bundle a broader set of responsibilities within jobs to offset increased coordination costs.

7 Changes to the Tacit Content of Work

The model predicts that automation should disproportionately substitute for more codifiable tasks, leaving the work performed by workers relatively more tacit. This prediction applies across automation technologies. I test this implication by examining how automation changes the tacit content of the tasks that remain with labor.

7.1 Composition of Remaining Human Work

I use two related specifications to examine automation's effects on tacit work. I first estimate how adoption of each automation technology affects the tacit share of the work

$$\theta_{T,ft} = \sum_{\tau} \beta_\tau \widehat{a}^{\tau}_{ft} + X'_{ft}\gamma + \lambda_{j(f),t} + \varepsilon_{ft}. \tag{30}$$

Here, $\theta_{T,ft}$ measures the tacit content of the work performed by firm f in year t. Higher values indicate that the firm's observed tasks rely more heavily on tacit knowledge.

I construct $\theta_{T,ft}$ using the Schaal codifiability scores introduced in Section 3. For each task d, I first transform its codifiability score into a continuous measure of tacitness:

$$\text{TacitScore}_d = 1 - \frac{\text{Codifiability}_d}{2}. \tag{31}$$

The transformation converts the codifiability measure into the unit interval, with higher values indicating greater reliance on tacit knowledge. I then aggregate these task scores across the tasks observed in firm f's job postings in year t to construct $\theta_{T,ft}$. The resulting firm-year measure also lies between zero and one and captures the average tacit content of the firm's observed work. An increase in $\theta_{T,ft}$ therefore indicates that the work remaining with workers has shifted toward tasks that rely more heavily on tacit knowledge.

Column (1) of Table 4 reports the estimates from Equation (30). The model predicts that automation should shift remaining human work toward more tacit tasks regardless of technology type, but the estimates provide only partial support for this prediction. Robotics and AI increase the average tacit content of the firm's tasks, with coefficients of 0.0336 and 0.0588. Software reduces tacit content however, with a coefficient of -0.0199. The compositional response to automation therefore differs across technologies: robotics and AI leave workers performing relatively more tacit work, while software moves the remaining task mix in the opposite direction.

The model predicts that automation is biased toward codified work on the displacement margin. I therefore examine whether the tasks that disappear following automation are disproportionately codified or tacit.

To compare these two types of work directly, I construct a stacked firm-year-by-task-type panel and estimate

$$\begin{aligned} Y^m_{fgt} = \sum_\tau \beta^m_\tau a^\tau_{ft} + \sum_\tau \phi^m_\tau \left(a^\tau_{ft} \times \text{Codified}_g\right) + \delta^m \text{Codified}_g + \lambda_{j(f),t} + \varepsilon^m_{fgt}, \\ m \in \{\text{rem}, \text{add}\}. \end{aligned} \tag{32}$$

Each firm-year contributes two observations to the stacked panel: one containing the outcome for tacit tasks and one containing the outcome for codified tasks. Thus, $g \in \{\text{tacit}, \text{codified}\}$ indexes task type within a firm-year. The outcome Y^m_{fgt} measures task removals when $m = \text{rem}$ and task additions when $m = \text{add}$. This design allows me to compare how the same automation treatment affects tacit and codified work within the same firm-year, rather than comparing these outcomes across different firms or years.

I classify tasks into the two groups using the Schaal codifiability measure. For task d, I define

$$\text{Tacit}_d = \mathbf{1}\left[\text{Codifiability}_d < 1\right], \qquad \text{Codified}_d = \mathbf{1}\left[\text{Codifiability}_d \geq 1\right]. \tag{33}$$

Tasks above this threshold form the codified group, while tasks below it form the tacit group. This discrete classification differs from the continuous tacit-content measure used above: the continuous measure captures changes in the average tacit content of the firm's work, whereas the binary classification allows me to compare task flows across the two sides of the codifiability threshold.

I define a task as removed when it was previously established at the firm but subsequently disappears from all future postings by the firm. To reduce the influence of infrequently observed tasks that may appear or disappear because of sampling variation, I classify a task as established only if it appeared in at least 50 percent of the firm's previously observed years.

To study task removals, I set $m = \text{rem}$ in Equation (32). When $g = \text{tacit}$, $\text{Codified}_g = 0$, so the effect of technology τ on tacit-task removals is β_τ^{rem}. When $g = \text{codified}$, $\text{Codified}_g = 1$, so the corresponding effect on codified-task removals is $\beta_\tau^{\text{rem}} + \phi_\tau^{\text{rem}}$. Thus I can construct

$$\phi_\tau^{\text{rem}} = \beta_{\text{codified},\tau}^{\text{rem}} - \beta_{\text{tacit},\tau}^{\text{rem}}. \tag{34}$$

The interaction coefficient ϕ_τ^{rem} is the coefficient of interest. A positive value indicates that automation removes codified tasks removals more than tacit ones, while a negative value indicates that the firms removes more tacit tasks than codified.

Column (2) of Table 4 reports the removal results. Contrary to the model's prediction that automation would disproportionately remove codified tasks, the estimated for robotics and AI are negative. The coefficient is -0.2016 for robotics and -0.9135 for AI, indicating that automation removes more tacit tasks than codified tasks. For software, the estimate is not distinguishable from zero. These findings create a tension with the results in column (1). Robotics and AI increase the tacit content of the work that remains with workers even though their displaced tasks are more tacit than codified. However, the composition of remaining work depends not only on which tasks disappear, but also on which new tasks appear. I next examine the addition of new tasks to determine whether the composition of newly appearing work helps reconcile these patterns.

I next examine the addition of new work following automation. In the model, automation can generate new complementary tasks that remain with workers. Examining the content of these additions helps determine whether newly created human work contributes to the changes in the tacit composition of work documented in column (1).

I define a task as an addition when it is absent from the firm's entire prior history of job postings, first appears in year t, and remains present in year $t + 1$. Requiring the task to persist into the following year reduces the likelihood that a task is classified as newly added because of transitory variation in the firm's postings.

To study additions, I set $m = \text{add}$ in Equation (32), so that Y_{fgt}^{add} measures newly added tasks of type g. The stacked structure and interpretation follow the removal analysis above. For tacit tasks, $\text{Codified}_g = 0$, so the effect of technology τ is β_τ^{add}. For codified tasks, $\text{Codified}_g = 1$, so the corresponding effect is $\beta_\tau^{\text{add}} + \phi_\tau^{\text{add}}$. Therefore,

Table 4: Effects of Automation on the Tacit Content of Work

	Average Tacit Score	log(Tacit Removals)	log(Tacit Additions)
	(1)	(2)	(3)
log(Robotics)	0.0336***	-0.2016***	-0.2191***
	(0.0011)	(0.0105)	(0.0094)
log(Software)	-0.0199*	0.0478	0.2001*
	(0.0103)	(0.1290)	(0.1200)
log(AI)	0.0588***	-0.9135***	-0.7131***
	(0.0058)	(0.0733)	(0.0646)
First-stage F: Robotics	3149.4	2990.4	2984.4
SW partial F: Robotics	3079.9	2694.3	2798.7
First-stage F: Software	2084.2	2517.6	2246.7
SW partial F: Software	130.8	101.6	123.1
First-stage F: AI	1510.2	1435.3	1870.6
SW partial F: AI	111.7	78.8	112.1
Observations	563,117	1,036,368	828,006
Firms	191,073	167,898	180,942
Controls	Yes	No	No
NAICS-4 $\times$ Year FE	Yes	Yes	Yes

Notes: Column (1)'s dependent variable is the average tacit score of tasks. Columns (2) and (3) report a stacked codified-vs-tacit differential: each firm-year contributes two rows to the estimation sample, one for its tacit tasks and one for its codified tasks, so the reported observation count is approximately twice the number of underlying firm-years. Column (1) includes controls for baseline average posting word count, lagged log employment, and lagged firm age. All specifications include four-digit NAICS-by-year fixed effects. Standard errors are clustered at the firm level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

$$\phi_{\tau}^{\text{add}} = \beta_{\text{codified},\tau}^{\text{add}} - \beta_{\text{tacit},\tau}^{\text{add}}. \tag{35}$$

A negative ϕ_{τ}^{add} indicates that automation adds tacit tasks more than it adds codified tasks, while a positive value indicates that newly added work consists of more codified tasks than tacit tasks.

Column (3) of Table 4 reports these results. Robotics and AI introduce more tacit additions. The coefficient is -0.2191 for robotics and -0.7131 for AI. Software instead produces a positive differential of 0.2001, indicating that introducing software automation adds more codified tasks than tacit tasks.

The results in column (3) help reconcile the net-composition results with the removal results for robotics and AI. The reason that both technologies could see a greater removal of tacit tasks while still seeing an increase in the tacit content of work is because additions were also dispproportionately tacit. So even though robotics and AI remove relatively more tacit work, the addition of new tacit tasks results in the remaining work becoming more tacit overall. Software exhibits a different pattern however: it shifts overall work toward greater codifiability, shows no change in removals, and demonstrates that the shift toward codified work stems from the addition of codified tasks. These results demonstrate that automation technologies differ in how they reshape the composition of human work.

8 Discussion

The results suggest that treating automation as a single technological treatment can conceal important differences in how firms reorganize work. Much of the task-based literature has studied automation by focusing on task displacement and task creation. Aggregating technologies can serve to identify these broad forces, but the organizational consequences of adoption might not be common across technologies. In the data, aggregate automation modestly broadens jobs, yet this average masks sharply different responses: robotics and software broaden jobs, while AI narrows them. These differences persist across alternative measures of job breadth, and the robotics and AI effects remain even when comparing firms hiring into the same occupation in the same year. The results therefore suggest that technology type matters not only for which tasks can be automated, but also for how firms reorganize the work that remains.

The findings also highlight a distinction between technological exposure and realized adoption. A growing literature measures the extent to which occupations or tasks are exposed to new technologies, providing evidence on where automation has the potential to affect work. This paper takes on a different approach. Firms reveal their inventory of automation investments through the skills appearing in their job postings. Exposure and adoption may not allows overlap. Firms facing similar technological exposures may differ in whether and when they introduce a technology into production. The measure I use here is imperfect, since demand for automation-related skills is not a direct observation of technology installation, but it illustrates the value of complementing exposure measures with measures that record firms' revealed investments in automation.

More broadly, the evidence suggests that job design is a component of the adjustment to technological change. Research on automation has naturally emphasized outcomes such as employment, wages, occupational composition, and the allocation of tasks between workers and machines. The results here show that firms can also respond by changing how the tasks allocated to labor are organized into jobs. The coordination evidence provides one possible explanation for this organizational response. The findings do not establish that coordination mediates the effect of automation on job breadth, but they are consistent with the idea that technologies differ in whether they create new coordination requirements or reduce the coordination required of workers.

Finally, the results also show that automation changes the tacit content of the work that remains with workers. A common intuition in the automation literature is that routine, codified work is more automatable. The results provide only partial support for this view. Robotics and AI leave the firm's remaining work more tacit, but software shifts it toward greater codifiability. Moreover, robotics and AI disproportionately remove tacit tasks even while increasing the tacit content of the work that remains. Their newly appearing tasks are also disproportionately tacit, showing that the post-adoption composition of human work depends on both the tasks that disappear and the new tasks that emerge around a technology. Taken together, these findings show that automation affects which tasks machines perform, which new tasks emerge, and how firms organize the work that remains.

9 Conclusion

This paper studies how firms redesign jobs when automation changes the work performed by workers. I extend the model of Becker and Murphy (1992), in which firms trade off the productivity gains from specialization against coordination costs, by allowing automation to change the composition of the work that remains. Automation changes the specialization problem by changing the tasks that remain with workers and the coordination required to divide those tasks across jobs. The empirical results show that the effects of automation differ depending on which technology is studied. Robotics and software broaden jobs, while

AI narrows them. These differences are also reflected in the coordination content of work. Robotics and software increase both coordination among workers and between workers and technology, whereas AI reduces both.

The composition of human work reveals a different pattern. Robotics and AI leave workers performing more tacit work, while software shifts work toward greater codifiability. Task flows further show that robotics and AI disproportionately remove tacit tasks while also generating tacit new work. The work left to humans after automation therefore cannot necessarily be inferred from the kinds of tasks that technology displaces. Automation simultaneously removes existing tasks and creates new ones.

Several limitations qualify these conclusions. Job postings describe positions firms seek to fill rather than the complete task content of incumbent workers, so the analysis is most informative about job design among firms with sufficiently active online hiring. Technology adoption is also inferred from firms' inventory of automation-related skills rather than directly observed technology installation. Finally, robotics, software, and AI each encompass substantial technological heterogeneity, and the estimates should be interpreted as average responses within these broad categories rather than effects shared by every technology they contain.

These limitations point toward several directions for future work. More direct measures of technology deployment, worker-level task histories, and internal personnel records could provide a closer view of how particular technologies redraw job boundaries. Future research may also benefit from distinguishing technologies according to the work they displace, the complementary work they create, and the coordination requirements they impose rather than treating automation as a single technological category. More broadly, the results suggest that organizational design should itself be viewed as an additional outcome of technological change.

This paper studies how firms redesign jobs when automation changes the tasks performed by workers. I develop a model in which firms choose how finely to divide human work by trading off the productivity benefits of specialization against the costs of coordinating tacit knowledge across workers. Automation changes this choice through two margins: the composition of remaining human work and the size of the human task set. Empirically, these forces produce sharply different organizational responses across technologies. Robotics and software broaden jobs, while AI narrows them, a pattern that tracks how each technology changes the coordination work assigned to workers: robotics and software increase the coordination required among workers and between workers and technology, while AI reduces both. Changes in the tacit content of remaining human work follow a different pattern: robotics and AI make that work more tacit, while software makes it less tacit. Because the compositional response does not line up with the job-design response across the three technologies, these results suggest that differences in how automation changes the coordination requirements of work, rather than differences in the direction of compositional change alone, are central to understanding why automation technologies reorganize jobs differently.

Several limitations qualify these conclusions. First, job postings measure the content of positions firms seek to fill rather than the complete set of tasks performed by incumbent workers. The analysis therefore measures job design through firms' hiring activity and is most informative for firms with sufficiently active online recruiting. Second, technology adoption is inferred from firms' accumulated demand for automation-related skills rather than observed purchases or deployment of particular technologies. Although the shift-share design addresses endogenous adoption under its exclusion restriction, it cannot eliminate every source of correlated change in firms' technology and organization. Third, the three technology categories summarize substantial heterogeneity within robotics, software, and AI, and the results should not be interpreted as implying that every technology within a category has the same effect.

These limitations suggest several directions for future research. One is to study more directly how particular technologies change job boundaries using worker-level task histories, internal personnel data, or detailed records of technology deployment. A second is to develop richer classifications of automation based on the human tasks technologies remove and create rather than treating automation as a single category. More generally, future work can treat organizational design as an outcome of technological change rather than as a fixed feature of production. Automation changes more than the allocation of tasks between humans and machines. It changes the set of human tasks that remain, the knowledge required to perform them, and ultimately the boundaries of the jobs into which firms organize human work.

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A Additional Details on Data and Variable Construction

This appendix provides additional detail on the construction of the variables used in the empirical analysis. The main text introduces each variable when it first becomes relevant; here I collect the implementation details needed to reproduce the measures of job breadth, task content, automation adoption, and the human task set.

A.1 Mapping Job Postings to Tasks

I measure the task content of jobs by mapping the text of Revelio Labs job postings to O*NET Detailed Work Activities (DWAs). Before constructing the firm-level measures used in the analysis, I restrict attention to firm-years with at least 10 observed job postings and at least three occupations. These restrictions reduce sensitivity to changes in job breadth generated by a small number of idiosyncratic vacancies.

For each posting, I compare the semantic content of its text with the descriptions of O*NET DWAs using the sentence-transformer model `all-MiniLM-L6-v2`. The model embeds the posting text and each candidate task description in a common vector space. I assign a DWA to a posting when the cosine similarity between the two embeddings is at least 0.45.

To reduce the influence of unusually long or generic postings that match a very large number of tasks, I cap the number of assigned DWAs at 70 per posting. When more than 70 DWAs exceed the similarity threshold, I retain the 70 tasks with the highest cosine similarity.

Let $\mathcal{D}_p$ denote the resulting set of DWAs assigned to posting p. Posting-level job breadth is

$$B_p = |\mathcal{D}_p|. \tag{36}$$

The main firm-year outcome averages this task count across the firm's postings:

$$\text{Breadth}_{ft} = \frac{1}{N_{ft}} \sum_{p \in (f,t)} |\mathcal{D}_p|, \tag{37}$$

where N_{ft} is the number of observed postings for firm f in year t. The regression outcome is $\log(\text{Breadth}_{ft})$.

This measure is intended to capture the breadth of the average posted job rather than the total variety of tasks performed somewhere in the firm. A higher value therefore corresponds to more tasks bundled into each job and, in the language of the model, lower specialization.

A.2 Task Codifiability

I measure the codifiability of human work using task-level codifiability scores constructed by Schaal (2025). Each DWA receives a score on a scale from zero to two, with higher values indicating greater codifiability.

The main composition analysis uses a continuous transformation of this measure. For DWA d, define

$$\text{TacitScore}_d = 1 - \frac{\text{Codifiability}_d}{2}. \tag{38}$$

The measure equals one for the least codifiable tasks and zero for the most codifiable tasks. I aggregate this measure across the tasks observed in the firm's postings to characterize the tacit content of human work.

The task-flow analysis instead requires a discrete classification because the outcomes count the entry and exit of individual task types. I classify a DWA as tacit when

$$\text{Codifiability}_d < 1 \tag{39}$$

and as higher codifiability when

$$\text{Codifiability}_d \geq 1. \tag{40}$$

The continuous and discrete measures therefore serve different empirical purposes. The continuous measure retains information about variation in codifiability within each category and is used for changes in the composition of remaining work. The discrete classification is used when counting gross task removal and reinstatement.

A.3 Technology-Specific Automation Measures

I distinguish three classes of automation technology: robotics, software, and AI. For each technology τ, I identify automation-related skills appearing in a firm's job postings and construct a cumulative firm-level stock of adoption.

Let A^{τ}_{ft} denote the cumulative stock of technology-τ adoption observed through year t. The treatment used in the regressions is

$$a^{\tau}_{ft} = \log\left(1 + A^{\tau}_{f,t-1}\right). \tag{41}$$

Lagging the stock ensures that the measured technology exposure precedes the outcome observed in year t. Years in which a firm has no observed adoption of technology τ are retained as zeros rather than treated as missing. This is important because absence of a technology-specific skill in a firm's postings represents zero measured adoption rather than an undefined technology stock.

A.4 Occupation-Based Shift-Share Instrument

Automation adoption may respond to changes in product demand, firm growth, workforce composition, or other factors that also affect job design. I therefore instrument firm adoption using differential exposure to national occupation-level adoption trends.

Let $\omega_{fo,0}$ denote occupation o's share of firm f's employment in the baseline period. For technology τ, let $G^{\tau}_{-f,-s(f),ot}$ denote the national change in technology adoption for occupation o in year t, excluding both the focal firm and firms located in the focal firm's state. The instrument is

$$Z^{\tau}_{ft} = \sum_{o} \omega_{fo,0} G^{\tau}_{-f,-s(f),ot}. \tag{42}$$

The leave-out construction prevents the focal firm's own adoption from mechanically entering its instrument. Excluding the focal state further reduces the possibility that the instrument captures contemporaneous local technology shocks or local labor-market conditions that directly affect the firm's organization.

Identification comes from the interaction of two sources of variation: predetermined differences in firms' occupational composition and national occupation-specific changes in technology adoption. Firms in the same industry and year can therefore receive different predicted technology shocks because their baseline workforces expose them differently to the same national trends.

B Alternative Measures of Job Breadth

The main analysis measures job breadth using O*NET Detailed Work Activities. This appendix examines whether the technology-specific results are sensitive to measuring work at broader levels of the O*NET hierarchy. I reconstruct job breadth using Intermediate Work Activities (IWAs) and Generalized Work Activities (GWAs) and re-estimate the preferred joint 2SLS specification from column (3) of Table 2, in which all three technology stocks enter simultaneously as endogenous regressors and are instrumented together.

Table 5 shows that the main technology-specific pattern is unchanged. Robotics and software broaden jobs under both alternative measures, while AI narrows them. The conclusion that different automation technologies reorganize jobs in different directions therefore does not depend on measuring job breadth using Detailed Work Activities.

Table 5: Alternative Measures of Job Breadth

	DWA Count (1)	IWA Count (2)	Element/GWA Count (3)
log(Robotics)	0.3529***	0.3182***	0.1223***
	(0.0114)	(0.0095)	(0.0070)
log(Software)	0.2245**	0.3321***	0.5114***
	(0.1032)	(0.0869)	(0.0707)
log(AI)	-0.6149***	-0.5666***	-0.4953***
	(0.0575)	(0.0486)	(0.0398)
First-stage F: Robotics	3149.1	3149.1	3149.1
SW partial F: Robotics	3079.7	3079.7	3079.7
First-stage F: Software	2084.2	2084.2	2084.2
SW partial F: Software	130.8	130.8	130.8
First-stage F: AI	1510.2	1510.2	1510.2
SW partial F: AI	111.7	111.7	111.7
Observations	563,115	563,115	563,115
Firms	191,071	191,071	191,071
Controls	Yes	Yes	Yes
NAICS-4 × Year FE	Yes	Yes	Yes

Notes: The dependent variable in column (1) is log average job breadth measured using O*NET Intermediate Work Activities; in column (2), using Generalized Work Activities. Both columns report the same joint 2SLS specification as column (3) of Table 2: all three technology stocks enter simultaneously as endogenous regressors and are instrumented together by their corresponding occupation-based shift-share instruments. Controls include baseline average posting word count, lagged log employment, and lagged firm age. All specifications include four-digit NAICS-by-year fixed effects. Standard errors, clustered at the firm level, are reported in parentheses. First-stage F is the own-instrument first-stage F-statistic for each technology's stock term; SW partial F reports the Sanderson-Windmeijer partial first-stage F, net of the identifying content already spanned by the other two instruments. Because both columns share the same estimation sample and first stage, First-stage F and SW partial F are identical across columns. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The sign pattern is stable across both alternative measures, and matches column (3) of Table 2 exactly: robotics and software increase job breadth, while AI decreases it, whether jobs are represented using Intermediate Work Activities or the broader Generalized Work Activities. The estimated effects generally become smaller as the task taxonomy becomes more aggregated, but the qualitative distinction across automation technologies remains unchanged.

C Occupation Classification and Automation Skill Dictionaries

This appendix documents the classification systems and keyword dictionaries used to construct the occupation and automation-technology measures in the main analysis.

C.1 O*NET Major Occupation Groups

Table 6 lists the 23 two-digit O*NET-SOC major occupation groups used to organize occupations throughout the paper.

Table 6: O*NET Major Occupation Groups

ONET SOC 2-digit	Name of Major Group
11	Management Occupations
13	Business and Financial Operations Occupations
15	Computer and Mathematical Occupations
17	Architecture and Engineering Occupations
19	Life, Physical, and Social Science Occupations
21	Community and Social Services Occupations
23	Legal Occupations
25	Educational Instruction and Library Occupations
27	Arts, Design, Entertainment, Sports and Media Occupations
29	Healthcare Practitioners and Technical Occupations
31	Healthcare Support Occupations
33	Protective Service Occupations
35	Food Preparation and Serving Related Occupations
37	Building and Grounds Cleaning and Maintenance Occupations
39	Personal Care and Service Occupations
41	Sales and Related Occupations
43	Office and Administrative Support Occupations
45	Farming, Fishing, and Forestry Occupations
47	Construction and Extraction Occupations
49	Installation, Maintenance, and Repair Occupations
51	Production Occupations
53	Transportation and Material Moving Occupations
55	Military Specific Occupations

C.2 Example Occupation Hierarchy

Within each major group, O*NET-SOC further nests occupations into minor groups, broad occupations, and detailed occupations. Table 7 illustrates this full hierarchy for major group

15 (Computer and Mathematical Occupations) as a representative example.

Table 7: Occupation Hierarchy within O*NET Category 15: Computer and Mathematical Occupations

Major Group	Minor Group	Broad Occupation	Detailed Occupation	Detailed SOC	Title
15-0000					Computer and Mathematical Occupations
	15-1200				Computer Occupations
		15-1210			Computer and Information Analysts
			15-1211		Computer Systems Analysts
				15-1211.01	Health Informatics Specialists
			15-1212		Information Security Analysts
		15-1220			Computer and Information Research Scientists
			15-1221		Computer and Information Research Scientists
		15-1230			Computer Support Specialists
			15-1231		Computer Network Support Specialists
			15-1232		Computer User Support Specialists
		15-1240			Database and Network Administrators
			15-1241		Computer Network Architects
				15-1241.01	Telecommunications Engineering Specialists
			15-1242		Database Administrators
			15-1243		Database Architects
				15-1243.01	Data Warehousing Specialists
			15-1244		Network and Computer Systems Administrators
		15-1250			Software and Web Developers
			15-1251		Computer Programmers
			15-1252		Software Developers
			15-1253		Software Quality Assurance Analysts and Testers
			15-1254		Web Developers
			15-1255		Web and Digital Interface Designers
				15-1255.01	Video Game Designers
			15-1299		Computer Occupations, All Other
				15-1299.01	Web Administrators
				15-1299.02	Geographic Information Systems Technologists and Technicians
				15-1299.03	Document Management Specialists
				15-1299.04	Penetration Testers
				15-1299.05	Information Security Engineers
				15-1299.06	Digital Forensics Analysts
				15-1299.07	Blockchain Engineers
				15-1299.08	Computer Systems Engineers/Architects
				15-1299.09	Information Technology Project Managers
	15-2000				Mathematical Occupations
		15-2010			Actuaries
			15-2011		Actuaries
		15-2020			Mathematicians
			15-2021		Mathematicians
		15-2030			Operations Research Analysts
			15-2031		Operations Research Analysts
		15-2040			Statisticians
			15-2041		Statisticians
				15-2041.01	Biostatisticians
		15-2050			Data Scientists
			15-2051		Data Scientists
				15-2051.01	Business Intelligence Analysts
				15-2051.02	Clinical Data Managers
			15-2099		Mathematical Science Occupations, All Other
				15-2099.01	Bioinformatics Technicians

C.3 Automation Skills List

Table 8 lists the Revelio Labs skill taxonomy entries used to identify automation-related skill references in job postings, prior to the technology-type partition described below.

Table 8: Automation-related skills list used to extract automation skill references in job postings

Automation skill
Artificial Intelligence
Automated Software Testing
Automation
Automation Testing
Build Automation
Building Automation
Business Process Automation
CNC
CNC Machine
CNC Manufacturing
CNC Mill
CNC Operation
CNC Programming
Computer Vision
Control System Design
Cucumber
DCS
Deep Learning
Distributed Control System
Distributed Control System (Dcs)
Fiddler
Hmi Design
Hmi Programming
Home Automation
Honeywell DCS
HP ALM
Human Machine Interface
Industrial Automation
Jmeter
Laboratory Automation
Machine Learning
Marketing Automation
Mes
Natural Language Processing
Nlp
PERL Automation
Pharmacy Automation
PLC
PLC Allen Bradley
PLC Ladder Logic
PLC Programming
PLC Siemens
Prism
Process Automation
Programmable Logic Controller
Programmable Logic Controller (Plc)
QA Automation
Robot
Robot Framework
Robot Programming
Robotics
Rslogix
SCADA
System Automation
Test Automation
Test Automation Framework
Test Automation Tools

C.4 Automation Keyword Dictionary by Technology Type

Table 9 reports the full keyword dictionary used to partition automation-related skill references into the four technology types (Robotics, AI, GenAI, and Software) used to construct the technology-specific adoption measures.

Table 9: Automation keywords used to extract automation skill references in job postings, partitioned by technology type.

Robotics	AI	GenAI (I)	GenAI (II)	Software	Software (II)
Building Automation	Computer Vision	agent framework	large language models	Automated Software Testing	PERL Automation
CNC	Deep Learning	agentic ai	llama	Automation	Prism
CNC Machine	Machine Learning	agentic workflow	llama 2	Automation Testing	Process Automation
CNC Manufacturing	Natural Language Processing	ai agent	llama index	Build Automation	QA Automation
CNC Mill	Nlp	ai agents	llama-3	Business Process Automation	System Automation
CNC Operation	***Additions***	ai alignment	llama-index	Cucumber	Test Automation
CNC Programming	AI-driven automation	ai assistant	llama2	Fiddler	Test Automation Framework
Control System Design	airflow	ai chatbot	llama3	Jmeter	Test Automation Tools
DCS	algorithmic automation	ai code generation	llamaindex	Marketing Automation	***Additions***
Distributed Control System (Dcs)	amazon sagemaker	ai copilot	llm	***Additions***	document classification
Hmi Design	auto ml	ai safety	llms	Abbyy	eliminate manual tasks
Hmi Programming	auto-ml	anthropic	lora	accounting automation	ELT automation
Home Automation	automate classification	anthropic api	microsoft autogen	Airflow	end-to-end automation
Honeywell DCS	automate decision making	autogen	microsoft copilot	Ansible	enterprise automation
Human Machine Interface	automate document processing	autogpt	midjourney	Appian RPA	ETL automation
Industrial Automation	automate forecasting	autonomous agent	mistral 7b	attended automation	finance automation
Laboratory Automation	automate image recognition	autonomous agents	mistral ai	automate	forms automation
Mes	automate NLP workflows	chain of thought	mistral-7b	automate end-to-end processes	HR automation
Pharmacy Automation	automate predictions	chatgpt	multi agent	automate manual processes	hyperautomation
PLC	automate recommendations	claude api	multi-agent	automate repetitive tasks	IaC
PLC Allen Bradley	automated machine learning	claude model	multimodal	automated	IDP
PLC Ladder Logic	automl	code generation	multimodal ai	automated data pipelines	implement automation across the enterprise
PLC Programming	autonomous systems	cohere api	multimodal model	automated testing	infrastructure as code
PLC Siemens	azure machine learning	conversational ai	openai	automating	infrastructure automation
Programmable Logic Controller	azure ml	crewai	openai api	Automation Anywhere	intelligent automation
Programmable Logic Controller (Plc)	bentoml	dall-e	parameter efficient fine	automation architecture	intelligent document processing
Robot	cognitive automation	dall-e 3	peft	automation center of excellence	invoice automation
Robot Framework	comet ml	dalle	pre-trained model	automation CoE	IT automation
Robot Programming	databricks	diffusion model	pretrained model	automation deployment	job scheduling
Robotics	decision automation	diffusion models	prompt engineering	automation enablement	Kafka pipelines
Rslogix	dnn	dspy	pytorch	automation engineering	Kofax
SCADA	feature engineering pipeline	embedding model	qlora	automation framework	Kubernetes
Additions	feature store	enrollment	rag	automation governance	Kubernetes automation
ABB Robotics	google vertex	few-shot	rag pipeline	automation implementation	Luigi
AGV	intelligent systems	fine tuning	rag-based	automation initiatives	Microsoft Power Automate
Allen-Bradley	kubeflow	fine-tuning	red teaming	automation integration	NICE RPA
AMR	metaflow	finetuning	retrieval augmented generation	automation optimization	OCR automation
automated guided vehicles	ML automation	foundation model	retrieval-augmented generation	automation pipeline	operational automation
autonomous mobile robots	ml ops	foundation models	rlhf	automation platform	operational efficiency
autonomous robots	mlflow	function calling	semantic kernel	automation roadmap	Pega RPA
cobots	mlops	gemini	semantic search	automation scalability	platform automation
control systems	model deployment	gen ai	speech synthesis	automation solutions	Playwright
factory automation	model monitoring	genai	stable diffusion	automation strategy	Power Automate
FANUC	model registry	generative ai	storage	automation systems	Prefect

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Table 9 (continued)

Robotics	**AI**	**GenAI (I)**	**GenAI (II)**	**Software**	**Software (II)**
HMI	model serving	github copilot	synthetic data generation	automation transformation	process re-engineering
industrial controls	neptune	google gemini	text embedding	back-office automation	process redesign
KUKA	OCR	gpt	text generation	batch automation	procurement automation
ladder logic	predictive automation	gpt-3	text to image	Blue Prism	Puppet
logistics automation	prefect	gpt-4	text to speech	bot development	records automation
manufacturing automation	ray serve	gpt3	text-to-image	bot orchestration	reduce human intervention
motion control	reinforcement learning	gpt4	text-to-speech	build automation solutions	regression automation
production automation	sagemaker	haystack	tool calling	call center automation	replace manual processes
Robot Operating System	seldon	hugging face	vector database	Chef	robotic process automation
robotic arms	speech recognition	huggingface	vector embedding	ci cd	RPA
robotic systems	torchserve	image captioning	vector search	claims automation	runbook automation
robotics engineering	transformer models	image generation	vector store	CloudFormation	scale automation initiatives
Rockwell Automation	triton inference server	image-text	vision language model	container orchestration	Selenium
ROS	vertex ai	instruction following	vision-language model	content extraction	service automation
Siemens automation	wandb	instruction tuning	visual language model	continuous delivery	shared services automation
warehouse automation	weights and biases	langchain	vlm	continuous deployment	Snowflake automation
	zenml	large language model	zero-shot	continuous integration	software bots
				contract automation	streaming automation
				customer service automation	supply chain automation
				Cypress	systems automation
				Dagster	task scheduling
				data orchestration	Terraform
				data pipeline automation	test frameworks
				Databricks automation	test orchestration
				dbt	text extraction
				deploy automation at scale	UiPath
				design automated workflows	unattended bots
				develop automation frameworks	underwriting automation
				DevOps	WinAutomation
				digital automation	workflow automation
				digital transformation	workflow orchestration
				digital workers	workflow redesign
				digitization	WorkFusion
				document automation	

C.5 Group 1 (Interpersonal) and Group 2 (Machine-Interface) DWA Lists

Table 10 lists the twenty-one Group 1 (interpersonal integrative work) and twenty-three Group 2 (machine-interface integrative work) O*NET Detailed Work Activities used to construct the outcomes in Table 3. DWA-to-group assignment was performed independently across six large language model classification passes, an adversarial review pass, and manual approval.

Table 10: Detailed Work Activities (DWAs) comprising Group 1 (interpersonal integrative work) and Group 2 (machine-interface integrative work).

Group 1: Interpersonal	**Group 2: Machine-Interface**
Translate information for others.	Monitor the performance of computer networks.
Interpret cultural or religious information for others.	Monitor computer system performance to ensure proper operation.
Interpret financial information for others.	Troubleshoot equipment or systems operation problems.
Explain engineering drawings, specifications, or other technical information.	Diagnose equipment malfunctions.
Liaise between departments or other groups to improve function or communication.	Determine production equipment settings.
Relay information between personnel.	Load materials into equipment for processing.
Communicate dining or order details to kitchen personnel.	Load materials into production equipment.
Signal others to coordinate vehicle movement.	Feed materials or products into or through equipment.
Signal others to coordinate work activities.	Mount attachments or tools onto production equipment.
Signal equipment operators to indicate proper equipment positioning.	Mount materials or workpieces onto production equipment.
Coordinate with external parties to exchange information.	Position raw materials on processing or production equipment.

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Table 10 (continued)

Group 1: Interpersonal	**Group 2: Machine-Interface**
Coordinate operational activities with external stakeholders.	Calibrate scientific or technical equipment.
Coordinate activities with suppliers, contractors, clients, or other departments.	Calibrate equipment to specifications.
Coordinate software or hardware installation.	Set equipment guides, stops, spacers, or other fixtures.
Coordinate project activities with other personnel or departments.	Enter commands, instructions, or specifications into equipment.
Coordinate shipping activities with external parties.	Program equipment to perform production tasks.
Plan employee work schedules.	Program robotic equipment.
Prepare staff schedules or work assignments.	Install programs onto computer or computer-controlled equipment.
Prepare activity or work schedules.	Resolve computer network problems.
Prepare employee work schedules.	Troubleshoot issues with computer applications or systems.
Assign duties or work schedules to employees.	Resolve computer software problems.
	Provide technical support for computer network issues.
	Convert data among multiple digital or analog formats.

D Coordination Tasks and the Headline Scope Result

Table 11: Removing Coordination and Interface DWAs from the Headline Scope Outcome

	Original	Interpersonal Removed	Interface Removed	Both Removed
	(1)	(2)	(3)	(4)
log(Robotics)	0.4307***	0.4367***	0.4213***	0.4273***
	(0.0110)	(0.0110)	(0.0109)	(0.0109)
log(Software)	0.5555***	0.5231***	0.5284***	0.4955***
	(0.1039)	(0.1033)	(0.1033)	(0.1027)
log(AI)	-0.6600***	-0.6516***	-0.6415***	-0.6328***
	(0.0591)	(0.0587)	(0.0586)	(0.0583)
First-stage F: Robotics	3660.9	3661.0	3661.0	3661.0
SW partial F: Robotics	3344.6	3344.8	3344.8	3344.8
First-stage F: Software	3158.5	3158.5	3158.5	3158.5
SW partial F: Software	114.1	114.2	114.2	114.2
First-stage F: AI	1749.4	1749.4	1749.4	1749.4
SW partial F: AI	87.7	87.7	87.7	87.7
Observations	729,198	729,202	729,202	729,202
Firms	250,131	250,135	250,135	250,135
Mean of dep. var.	2.8666	2.8585	2.8647	2.8557
Controls	No	No	No	No
NAICS-4 × Year FE	Yes	Yes	Yes	Yes

Notes: Pooled (joint) 2SLS: all three technology stocks enter simultaneously as endogenous regressors and are instrumented together by their corresponding occupation-based shift-share instruments. Column (1) reproduces the headline specification with no tasks removed; columns (2)–(4) recompute average task count after dropping Group 1 (worker–worker coordination), Group 2 (worker–technology coordination), or both sets of DWAs from every posting's task list, respectively, before re-estimating the same specification. No controls (potential colliders). First-stage F is the own-instrument first-stage F-statistic for each technology's stock term; SW partial F reports the Sanderson-Windmeijer partial first-stage F, net of the identifying content already spanned by the other two instruments. Standard errors, clustered at the firm level, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.